Iris-Based Intent Selection
for the VLM Robotic Arm Demo
(as an EEG Alternative with Minimal Code Changes)

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1. Summary

The VLM robotic arm demo was originally planned to use an EEG device for intent selection (choosing which object to grab). Because the EEG device will not be available for the 28 Jan 2026 demo, this proposal replaces only the intent-selection block with an iris-based module. The core message is simple: the perception pipeline (VLM detection), the downstream state machine, and the arm control code stay the same. Only the input signal that decides "which target did the user mean" is swapped.

This approach preserves the "telepathic" feel of the demo: the user does not click or type. The system flashes candidates on screen, reads the user's eye response, and produces a single target ID plus a trigger signal, just like the EEG design. The prototype is already working end-to-end and uses common hardware, which makes it highly achievable for a live demo.

2. Background

The system has three parts. First, perception: the camera and VLM pipeline detect and label objects. Second, intent selection: the user selects one target from the detected objects. Third, actuation: the robotic arm and gripper execute the approach and grasp sequence.

EEG was planned for the intent selection step. Without EEG, perception and actuation still run, but the system loses the hands-free target-selection experience. This proposal restores that experience by replacing only the intent-selection block, without changing the rest of the pipeline.

3. Goals

The demo has three priorities. First, minimal change: keep the existing VLM detection and robotic arm control logic intact. Second, WOW factor: keep the interaction feeling intention-driven, where the system appears to “read the user’s choice” without manual input. Third, achievability: use hardware and software that can be deployed reliably and quickly in a live demo setting.

4. Proposed solution

The proposed solution is an iris-based intent selection module using a standard RGB camera and MediaPipe Face Mesh. Candidates are displayed on screen and flashed in a controlled sequence. The system tracks the user’s left eye in real time and measures a pupil-response signal during each flash.

In the current implementation, the response signal is a normalized pupil metric derived from pupil area divided by eye width squared, which helps reduce sensitivity to distance changes and small camera zoom effects. This signal is used to score each candidate across multiple flashes and rounds, then select the best-matching target.

A short benchmark step runs before the selection loop to estimate the user’s typical response magnitude and to set a dynamic threshold. This makes the final decision more stable and reduces false selections in front of an audience.

5. Current status quo

A working prototype for the module is already written It automatically calibrates a camera threshold for pupil segmentation, continuously tracks the normalized pupil signal, detects blink and skips blink-affected samples, and records per-item response scores during the flash sequence. After the run, it selects the strongest candidate if the response exceeds the dynamic threshold; otherwise, it reports an inconclusive result and restarts.

Most importantly for integration, the module can output the same selection fields that the downstream controller expects: a selected target identifier and a trigger event. That means the stimulus and selection code does not need structural changes.

6. Minimal-change integration approach

The iris module is treated as a drop-in replacement for the EEG module. The interface between intent selection and actuation is kept identical. The VLM object list, the state machine, the motion planning logic, and the gripper control remain unchanged.

In practice, this is an adapter swap. Where the system previously waited for an EEG-based "selection" message, it now listens to an iris-based "selection" message in the same schema. Switching back to EEG later is also straightforward because the rest of the software stays untouched.

7. Wow effect

The audience experience is designed to feel like intention detection rather than manual control. The user only looks at the screen while the system flashes options. The selection happens without the user clicking, typing, or verbally confirming the target.

This creates a clean narrative for the demo: the VLM identifies objects, the system presents choices, and the user's "intent" is inferred through eye response. Even though the sensor is a camera instead of EEG, the visible behavior still matches the idea of mind-like control from the audience perspective.

8. Achievability

This approach is achievable because it does not depend on specialized lab hardware. It runs with a common webcam and a lightweight landmark tracker. It also avoids retraining a complex model by using a simple calibration step and a dynamic threshold derived from the user's own baseline responses.

9. Risks and mitigations

Iris and pupil tracking can degrade under poor lighting, glare from glasses, or large head movements. Mitigations include conservative thresholds, skipping blink-affected flashes, confidence gating when tracking quality drops, and a quick re-calibration path if conditions change.

10. Deliverables

Deliverables include the integrated iris intent module, a short demo recording, timestamped selection logs, and a short reliability summary (selection accuracy, false triggers, and end-to-end latency under demo-like conditions).

11. Conclusion

Replacing EEG with iris-based intent selection is the fastest way to keep the intended hands-free, telepathic-style demo . It preserves the existing software architecture and requires only an adapter-level change at the intent-selection interface, while still delivering a strong wow effect.

Because it uses readily available hardware and a working prototype already exists, this solution is highly achievable for a reliable live demonstration.